out by nitrifying bacteria. It is a two step process where ammonia is first converted to nitrite (NO_2) and then to nitrate (NO_3).

$$2NH_3 + 3O_2 \rightarrow 2HNO_2 + 2H_2O$$

The above given oxidation reaction is the first step that produces nitrite. This reaction provides energy and is carried out by *Nitrosomonas* and *Nitrosococcus*.

In the second step, the nitrite is oxidized to nitrate

$$2HNO_2 + O_2 \rightarrow HNO_3$$

This reaction is carried out by *Nitrobacter*.

Nitrates are readily assimilated by plants but are very water soluble and rapidly leached from soil.

Denitrification

The reduction of NO_3^- from soils by denitrifying bacteria to gaseous nitrogen is called **denitrification**. In this process, carried out by bacteria like *Pseudomonas, Thiobacillus denitrificans,* organic compounds serve as hydrogen donors and nitrates serve as electron acceptor.

$$NO_3 \rightarrow NO_2 \rightarrow N_2O \rightarrow N_2$$

Biological Nitrogen Fixation

One of the most significant biological process taking places on the Earth is biological nitrogen fixation (BNF). This fixation of atmospheric nitrogen carried out by few prokaryotes is cost efficient because industrial production of ammonia by Haber's Bosch process is very expensive.

Organisms carry out BNF in a free living state in soil or they can establish symbiotic association with other plants or micro organisms and fix N_2.

Organisms capable of BNF are

Free living	
Aerobic	*Azotobacter* Cyanobacteria
Anaerobic	*Clostridium*
Symbiotic	*Rhizobium* *Frankia.*

BNF by *Rhizobium* in leguminous plants

Leguminous plants belong to the family Leguminaceae and bear seeds in pods. Example: Black gram, Green peas, Soyabean, Subabul. The bacteria belonging to the genus *Rhizobium* which can exist in free living state in soil but can enter into symbiosis with legume plants and carry out nitrogen fixation.

Process of BNF

It consists of the following steps

1. Infection of legume roots by *Rhizobium*
2. Formation of root nodules
3. Reduction of $N \equiv N$ to NH_4 in root nodules

1. Infection of legume roots by *Rhizobium*

Rhizobium recognises and attaches to the root hairs of legume plant. It invades the root hairs and secretion of certain *nod* factors result in root hair curling typically called **Shepherd's crook symptom** which leads to the formation of infection thread. Infection thread is a cellulosic tube like structure through which *Rhizobium* moves into the cortex from root hairs.

2. Formation of root nodules

The invaded plant cells are stimulated to divide repeatedly thus forming a mass

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